

Linux networking commands

1. `ifconfig`:

- Description: Displays or configures network interface parameters for a network using the Internet Protocol. It shows information about network interfaces such as IP addresses, MAC addresses, and network statistics.

```
c/Users/paras took 3s
> ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1400
    inet 172.28.205.166 netmask 255.255.240.0 broadcast 172.28.207.255
    inet6 fe80::215:5dff:feed:911a prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:ed:91:1a txqueuelen 1000 (Ethernet)
    RX packets 256 bytes 229070 (229.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 106 bytes 7230 (7.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 34 bytes 3862 (3.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 34 bytes 3862 (3.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. `ip`:

- Description: A versatile tool for network management that replaces older tools like `ifconfig`. It is used to show and manipulate routing, devices, policy routing, and tunnels.

```
c/Users/paras
> ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:ed:91:1a brd ff:ff:ff:ff:ff:ff
    inet 172.28.205.166/20 brd 172.28.207.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:feed:911a/64 scope link
        valid_lft forever preferred_lft forever
```

3. `traceroute`:

- Description: Shows the route packets take from one host to another. It lists each hop along the path and helps in diagnosing network routing issues.

```

c/Users/paras took 3s
> traceroute google.com
traceroute to google.com (142.250.183.110), 30 hops max, 60 byte packets
 1  172.28.192.1 (172.28.192.1)  0.577 ms  0.541 ms  0.531 ms
 2  192.168.145.157 (192.168.145.157)  3.754 ms  3.579 ms  5.273 ms
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *

```

4. `tracert`:

- Description: Similar to `tracert`, it traces the network path to a target host and displays the route packets take, including any network issues or packet loss.

```

c/Users/paras took 35s
> tracert google.com
1?: [LOCALHOST] pmtu 1400
 1:  ??? 0.701ms
 1:  ??? 0.446ms
 2:  192.168.145.157 39.801ms
 3:  192.168.145.157 6.206ms pmtu 1372
 3:  no reply
 4:  no reply
 5:  no reply
 6:  no reply
 7:  no reply
 8:  no reply
 9:  no reply
10:  no reply
11:  no reply
12:  no reply
13:  no reply
14:  no reply
15:  no reply
16:  no reply
17:  no reply

```

5. `ping`:

- Description: Sends ICMP Echo Request messages to a host to check its reachability and measure round-trip time. It's useful for checking if a network device is reachable and how long it takes to communicate.

```
c/Users/paras took 1m32s
> ping google.com
PING google.com (142.251.42.14) 56(84) bytes of data.
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=1 ttl=110 time=110 ms
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=2 ttl=110 time=120 ms
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=3 ttl=110 time=143 ms
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=4 ttl=110 time=185 ms
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=5 ttl=110 time=85.4 ms
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=6 ttl=110 time=200 ms
```

6. netstat:

- Description: Displays network connections, routing tables, interface statistics, masquerade connections, and multicast memberships. It's useful for network troubleshooting and monitoring.

```
c/Users/paras took 16s
> netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.53:domain      0.0.0.0:*               LISTEN
tcp        0      0 10.255.255.254:domain  0.0.0.0:*               LISTEN
udp        0      0 127.0.0.53:domain      0.0.0.0:*
udp        0      0 10.255.255.254:domain  0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 ip6-localhost:323      [::]:*
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State       I-Node     Path
unix   2      [ ACC ] STREAM    LISTENING   20519      /run/WSL/2_interop
unix   2      [ ACC ] STREAM    LISTENING   18722      /run/WSL/1_interop
unix   2      [ ACC ] STREAM    LISTENING   19510      /var/run/dbus/system_bus_socket
unix   2      [ ACC ] SEQPACKET LISTENING   17433      /mnt/wslg/weston-notify.sock
unix   2      [ ACC ] STREAM    LISTENING   153       /mnt/wslg/runtime-dir/wayland-0
```

7. nslookup:

- Description: Queries DNS to obtain domain name or IP address mapping. It helps in diagnosing DNS-related issues by providing details about DNS records.

```
c/Users/paras
> nslookup google.com
Server:      10.255.255.254
Address:     10.255.255.254#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.42.14
Name:   google.com
Address: 2404:6800:4009:823::200e
```

8. dig:

- Description: A flexible tool for querying DNS nameservers to retrieve domain name information. It provides detailed information about DNS records and is often used for troubleshooting DNS issues.

```

c/Users/paras
> dig google.com

; <<>> DiG 9.18.18-0ubuntu0.22.04.1-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 28235
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 115     IN      A      142.251.42.14

;; Query time: 70 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Mon Sep 09 00:37:07 IST 2024
;; MSG SIZE rcvd: 44

```

9. `route`:

- Description: Displays or modifies the IP routing table. It is used to view or configure routes for network traffic and can help in diagnosing routing issues.

```

c/Users/paras
> route -n
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0        172.28.192.1   0.0.0.0         UG    0      0      0 eth0
172.28.192.0   0.0.0.0        255.255.240.0   U      0      0      0 eth0

```

10. `host`:

- Description: Performs DNS lookups to resolve domain names to IP addresses and vice versa. It's a simpler alternative to `dig` for querying DNS information.

```

c/Users/paras
> host google.com
google.com has address 142.251.42.14
google.com has IPv6 address 2404:6800:4009:823::200e
google.com mail is handled by 10 smtp.google.com.

```

11. `arp`:

- Description: Displays and modifies the ARP (Address Resolution Protocol) cache. This cache maps IP addresses to MAC addresses on a local network.

```

c/Users/paras took 7s
> arp -a
? (172.28.192.1) at 00:15:5d:f1:07:88 [ether] on eth0

```

12. `iwconfig`:

- Description: Configures wireless network interfaces. It provides information about the wireless connection, such as signal strength and access points.

```
c/Users/paras took 6s
> iwconfig
lo          no wireless extensions.

eth0        no wireless extensions.
```

13. `curl`:

- Description: Transfers data from or to a server using various protocols (HTTP, HTTPS, FTP, etc.). It's used for interacting with web services and APIs.

```
c/Users/paras
> curl http://example.com
<!doctype html>
<html>
<head>
  <title>Example Domain</title>

  <meta charset="utf-8" />
  <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <style type="text/css">
    body {
      background-color: #f0f0f2;
      margin: 0;
      padding: 0;
      font-family: -apple-system, system-ui, BlinkMacSystemFont, "Segoe
s-serif;

    }
    div {
      width: 600px;
      margin: 5em auto;
      padding: 2em;
      background-color: #fdfdff;
      border-radius: 0.5em;
      box-shadow: 2px 3px 7px 2px rgba(0,0,0,0.02);
    }
    a:link, a:visited {
      color: #38488f;
      text-decoration: none;
    }
  </style>
</head>
<body>
  <div>
    <h1>Example Domain</h1>
    <p>This domain is for use in illustrative examples in documents. You
may use this domain to illustrate network configurations that do not
require Internet access. This domain is not suitable for use in
educational institutions, companies or domains that require Internet
access.
    </p>
  </div>
</body>
</html>
```

14. `wget`:

- Description: A command-line utility for downloading files from the web. It supports HTTP, HTTPS, and FTP protocols and can download multiple files and directories recursively.

```
c/Users/paras
> wget http://example.com/file.zip
--2024-09-09 00:41:32-- http://example.com/file.zip
Resolving example.com (example.com)... 93.184.215.14, 2606:2800:21f:cb07:6820:80da:af6b:8b2c
Connecting to example.com (example.com)|93.184.215.14|:80... connected.
HTTP request sent, awaiting response... 404 Not Found
2024-09-09 00:41:33 ERROR 404: Not Found.
```

15. `telnet`:

- Description: Provides a command-line interface for communicating with a remote host using the Telnet protocol. It is used for testing and debugging network services.

```
c/Users/paras took 6s
> telnet example.com 80
Trying 93.184.215.14...
Connected to example.com.
Escape character is '^]'.
GET / HTTP/1.1
Host: example.com

HTTP/1.1 200 OK
Accept-Ranges: bytes
Age: 261718
Cache-Control: max-age=604800
Content-Type: text/html; charset=UTF-8
Date: Sun, 08 Sep 2024 19:24:34 GMT
Etag: "3147526947+gzip"
Expires: Sun, 15 Sep 2024 19:24:34 GMT
Last-Modified: Thu, 17 Oct 2019 07:18:26 GMT
Server: ECAcc (nyd/D10B)
Vary: Accept-Encoding
X-Cache: HIT
Content-Length: 1256

<!doctype html>
<html>
<head>
  <title>Example Domain</title>

  <meta charset="utf-8" />
  <meta http-equiv="Content-type" content="text/html; charset=utf
  <meta name="viewport" content="width=device-width, initial-scal
```

16. `whois`:

- Description: Queries domain registration information from a WHOIS database. It provides details about domain ownership, registration, and contact information.

```

c/Users/paras took 6s
) whois example.com
Domain Name: EXAMPLE.COM
Registry Domain ID: 2336799_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.iana.org
Registrar URL: http://res-dom.iana.org
Updated Date: 2024-08-14T07:01:34Z
Creation Date: 1995-08-14T04:00:00Z
Registry Expiry Date: 2025-08-13T04:00:00Z
Registrar: RESERVED-Internet Assigned Numbers Authority
Registrar IANA ID: 376
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: A.IANA-SERVERS.NET
Name Server: B.IANA-SERVERS.NET
DNSSEC: signedDelegation
DNSSEC DS Data: 370 13 2 BE74359954660069D5C63D200C39F5603827D7DD02B56F120EE9F3A86764247C
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2024-09-08T19:27:59Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration

```

17. `ifplugstatus`:

- Description: Displays the status of network interfaces and their connection state. It is useful for checking if an Ethernet cable is plugged in or not.

```

c/Users/paras took 6s
) ifplugstatus
lo: link beat detected
eth0: link beat detected

```

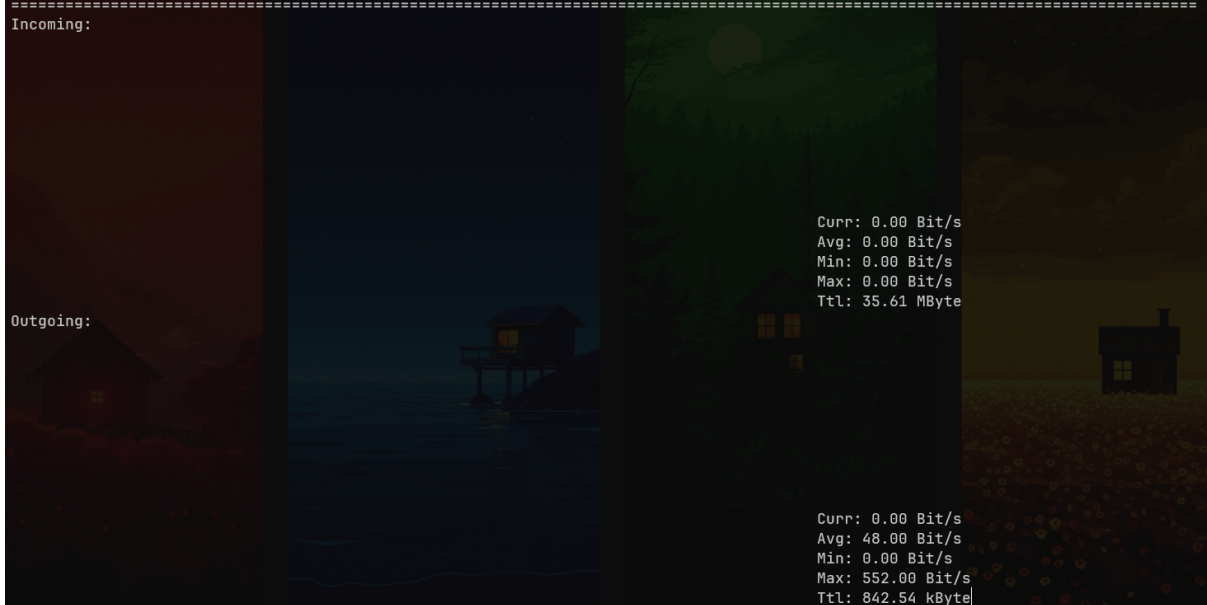
18. `nload`:

- Description: Monitors network traffic and bandwidth usage in real-time. It provides a graphical representation of incoming and outgoing traffic on network interfaces.

```

Device eth0 [172.28.205.166] (1/2):
=====
Incoming:

```



```

Outgoing:

```

```

Curr: 0.00 Bit/s
Avg: 48.00 Bit/s
Min: 0.00 Bit/s
Max: 552.00 Bit/s
Ttl: 842.54 kByte

```

19. mail:

- **Description:** A command-line utility for sending and receiving email. It is used for composing, sending, and managing email messages from the terminal.

```
((base) yashpardeshi@Yashs-MBP ~ % sudo postfix start
Password:
postfix/postfix-script: fatal: the Postfix mail system is already running
((base) yashpardeshi@Yashs-MBP ~ % echo "This is the body of the email" | mail -s "Test Subject" recipient@example.com
(base) yashpardeshi@Yashs-MBP ~ % tail -f /var/log/system.log
Oct 3 11:00:39 Yashs-MBP syslogd[332]: ASL Sender Statistics
Oct 3 11:11:07 Yashs-MBP syslogd[332]: ASL Sender Statistics
Oct 3 11:13:02 Yashs-MBP login[1552]: USER_PROCESS: 1552 ttys002
Oct 3 11:22:07 Yashs-MBP login[1552]: DEAD_PROCESS: 1552 ttys002
Oct 3 11:22:07 Yashs-MBP syslogd[332]: ASL Sender Statistics
Oct 3 11:35:44 Yashs-MBP syslogd[332]: ASL Sender Statistics
Oct 3 11:43:12 Yashs-MBP login[3611]: USER_PROCESS: 3611 ttys002
Oct 3 11:45:13 Yashs-MBP login[3611]: DEAD_PROCESS: 3611 ttys002
Oct 3 11:46:07 Yashs-MBP syslogd[332]: ASL Sender Statistics
Oct 3 11:48:45 Yashs-MBP login[4866]: USER_PROCESS: 4866 ttys002
```

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